CSS

Cascading Style Sheets

CSS

- **Cascading Style Sheets** (**CSS**): is a simple mechanism for *adding style* (e.g. fonts, colors, layouts) to Web documents.
- Styles provide *powerful control* over the **presentation** of web pages



What is CSS?

CSS stands for “Cascading Style Sheets”

Cascading: refers to the procedure that determines which style will apply to a certain section, if you have more than one style rule.

What is CSS?

CSS stands for “Cascading Style Sheets”

Style: how you want a certain part of your page to look. You can set things like color, margins, font, etc for things like tables, paragraphs, and headings.

What is CSS?

CSS stands for “Cascading Style Sheets”

Sheets: the “sheets” are like templates, or a set of rules, for determining how the webpage will look.

So, CSS (all together) is a *styling language* – a set of rules to tell browsers how your webpage should look.

What is “Style”

“Style” is a command that you set to tell the browser how a certain section of your webpage should look.

You can use style on many HTML “elements” (like <p> <h1> <table> etc)

Enter CSS Timeline

CSS1- December 1996

CSS 2- Became a W3C Recommendation in May 1998

CSS 3- CSS level 3 has been under development since December 15, 2005

CSS Versions

- **CSS 1**
  - Font properties such as typeface and emphasis
  - Color of text, backgrounds, and other elements
  - Text attributes such as spacing between words, letters, and lines of text
  - Alignment of text, images, tables and other elements
  - Margin, border, padding, and positioning for most elements
  - Unique identification and generic classification of groups of attributes
- CSS2

  includes a number of new capabilities like
  - absolute, relative, and fixed positioning of elements and z-index,
  - the concept of media types
  - support for aural style sheets and bidirectional text
  - new font properties such as shadows.

CSS Versions

- CSS3

  Modules include:
  - **Borders (**border-radius, box-shadow)
  - **Backgrounds (m**ultiple backgrounds)
  - **Color (**HSL colors, HSLA colors, opacity, RGBA colors)
  - Also:
  - media queries
  - multi-column layout
  - Web fonts

The Separation of Style, Content and Behavior

- Core Concept of web development
- HTML + CSS + JavaScript
- Content + Style + Behavior

How to write style rules

Two parts: (1) selector and (2) declaration.

Selector: the HTML element you want to add style to.

<p> <h1> <table> etc

Declaration: the statement of style for that element. Made up of *property* and *value*.

How to write style rules

Selector {declaration;}

Declaration = {property: value;}

Property: what aspect you want to change

ex: color, font, margins, etc.

Value: the exact setting for that aspect.

ex: red, italic, 40px, etc.

How to write style rules

selector {property: value;}

Essentially means:

The thing I want to change

{the aspect of that thing I want to change:
what I want it to be;}

How to write style rules

Selector {property: value;}

h1 {color: red;}

Means: I want the text color of my heading1 to be red.

How to write style rules

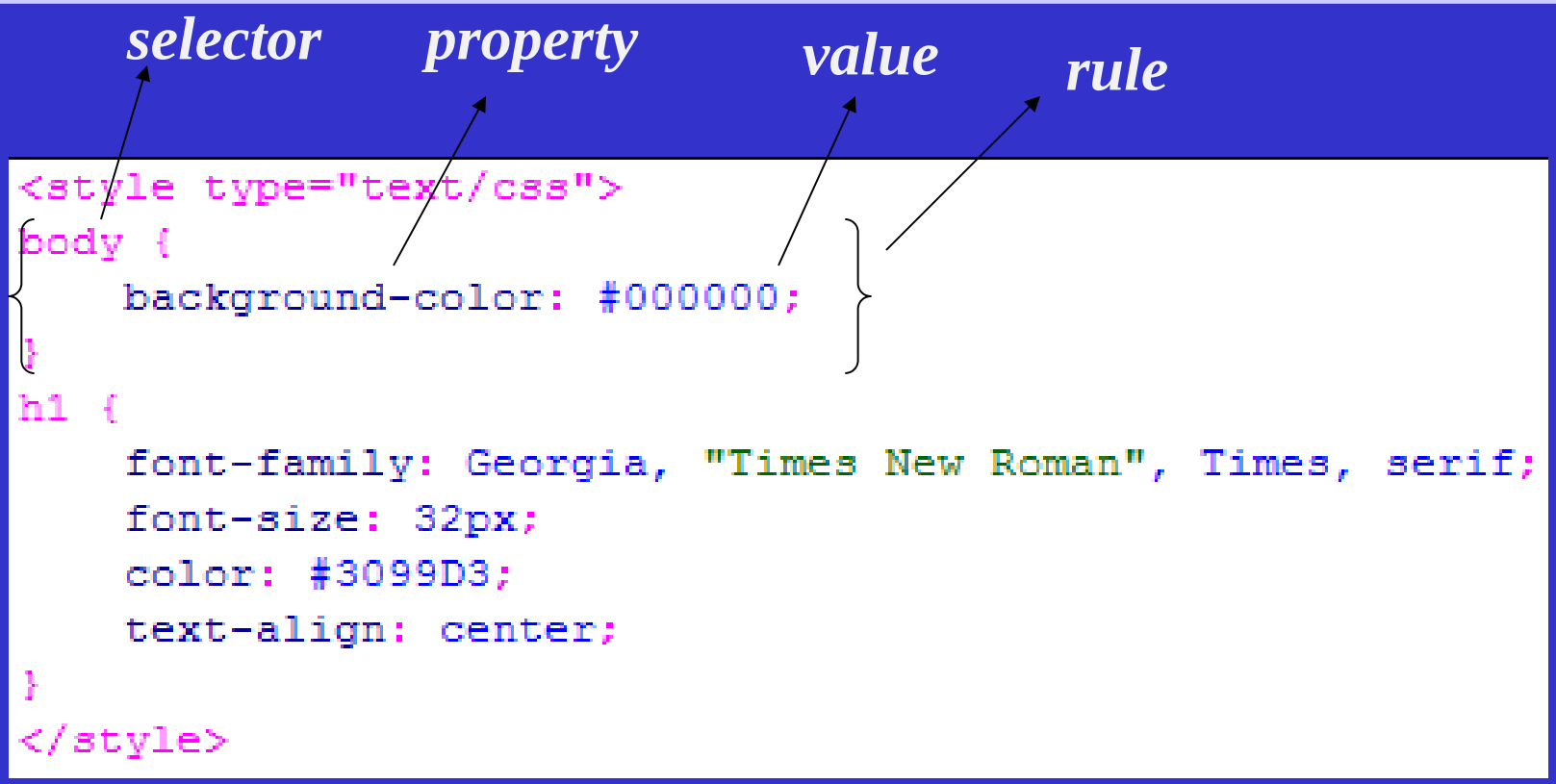
Selector {property: value;}

I want the color of the text in my paragraph to be yellow.

P{color: yellow;}

This is the text in this paragraph.

CSS


selector
property
value
rule
<style type="text/css">
body {
background-color: #000000;
}
h1 {
font-family: Georgia, "Times New Roman", Times, serif;
font-size: 32px;
color: #3099D3;
text-align: center;
}
</style>

CSS

- A styled HTML document



produced by the style sheet `style1.css`:

```
body   { background-color:lime }
p      { font-size:x-large; background-color:yellow }
```

CSS:

```
body {
    font-family: Tahoma, Arial, sans-serif;
    font-size: 13px;
    color: black;
    background: white;
    margin: 8px;
}
h1 {
    font-size: 19px;
    margin-top: 15px;
    margin-bottom: 5px;
    border-bottom: 1px solid black
}
.shaded {
    background: #d0d0ff;
}
```

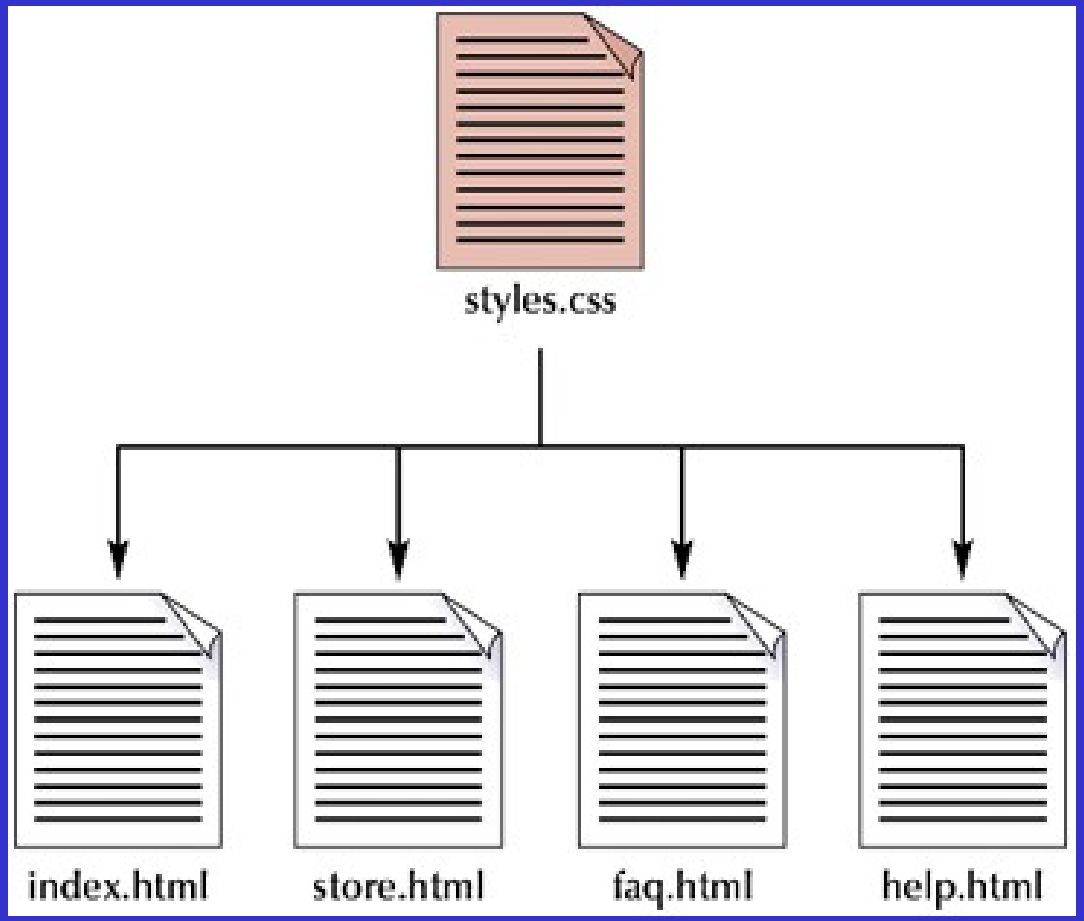
HTML

```
<body>
    <h1>First Section Heading</h1>
    <p>
        Here is the first paragraph, containing
        text that really doesn't have any use
        or meaning; it just prattles on and on,
        with no end whatsoever, no point to
        make, really no purpose for existence
        at all.
    </p>
    <div class="shaded">
        <h1>Another Section Heading</h1>
        <p>
          Another paragraph.
        </p>
    </div>
</body>
```

Three Different Scopes of CSS

- Local
  - confined to a **single element (tag)**
- Internal
  - affect elements in an **entire page**
- External
  - can affect **multiple pages**
- **Precedence**
  - **Local > Internal > External**

Applying a single style sheet to multiple documents



a) Inlined example-1

```
<html>
<head><title>My Wonderful Example</title>
</head>
<body>
        <p>What was I thinking?</p>
</body>
</html>
```



a) Inlined example-2

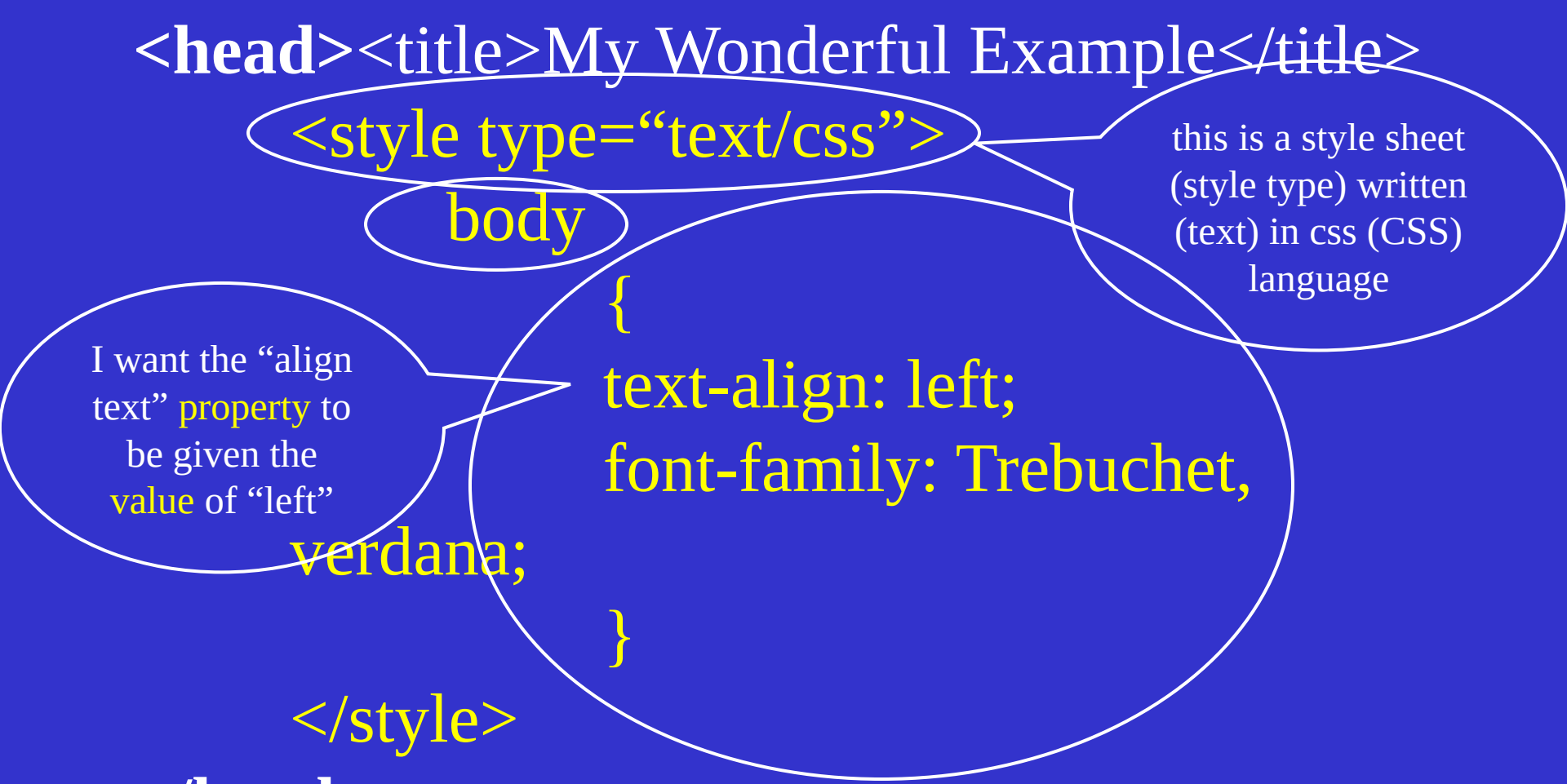
```
<body>
        <p style="text-align: center; font-
  weight: bold; color: yellow;">What was
  I thinking?</p>
</body>
```

What was I thinking?

b) Embedded/Internal-1

```
<head><title>My Wonderful Example</title>
        <style type="text/css">
                body
                        {text-align: left;
                        font-family: trebuchet,
        verdana;}
        </style>
</head>
```

b) Embedded/Internal-2



c) External

```
<html>
<head><title>My Way</title>
        <link rel="stylesheet"
  href="http://www2.hawaii.edu/~myway.
  css"   type="text/css“>
</head>
<body> </body>
</html>
```



CSS

```
<!DOCTYPE html
        PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      CSSHelloWorld.html
    </title>
    <link rel="stylesheet" type="text/css" href="style1.css"
          title="Style 1" />
    <link rel="alternate stylesheet" type="text/css" href="style2.css"
          title="Style 2" />
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

`link` element associates style sheet with doc.

CSS

```
<!DOCTYPE html
        PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      CSSHelloWorld.html
    </title>
    <link rel="stylesheet" type="text/css" href="style1.css"
          title="Style 1" />
    <link rel="alternate stylesheet" type="text/css" href="style2.css"
          title="Style 2" />
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

type attribute specifies style language used

CSS

```
<!DOCTYPE html
        PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      CSSHelloWorld.html
    </title>
    <link rel="stylesheet" type="text/css" href="style1.css"
          title="Style 1" />
    <link rel="alternate stylesheet" type="text/css" href="style2.css"
          title="Style 2" />
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

`href` attribute provides style sheet URL

CSS

```
<!DOCTYPE html
        PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      CSSHelloWorld.html
    </title>
    <link rel="stylesheet" type="text/css" href="style1.css"
          title="Style 1" />
    <link rel="alternate stylesheet" type="text/css" href="style2.css"
          title="Style 2" />
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

`title` attribute provides style sheet name

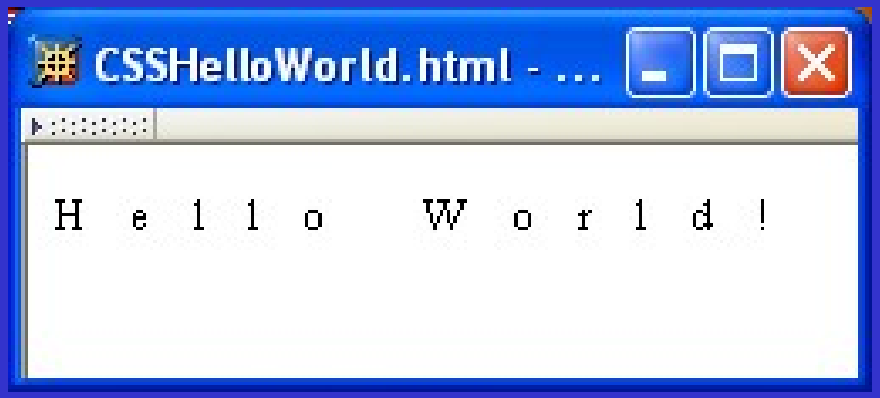
CSS

```
<!DOCTYPE html
        PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      CSSHelloWorld.html
    </title>
    <link rel="stylesheet" type="text/css" href="style1.css"
          title="Style 1" />
    <link rel="alternate stylesheet" type="text/css" href="style2.css"
          title="Style 2" />
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

Alternative, user selectable style sheets can be specified

CSS

- A styled HTML document



produced by the style sheet `style2.css`:

```
p      { font-size:smaller; letter-spacing:1em }
```

CSS Syntax: Selector Strings

- Source anchor elements:

```
a:link { color:black }
a:visited { color:yellow }
a:hover { color:green }
a:active { color:red }
```

pseudo-classes

- Element types that are descendents:

```
ul ol li { letter-spacing:1em }
```

CSS Syntax: Selector Strings

- Source anchor elements:

```
a:link { color:black }
a:visited { color:yellow }
a:hover { color:green }
a:active { color:red }
```

- Element types that are descendants:

```
ul ol li { letter-spacing:1em }
```

rule applies to `li` element that is

CSS Syntax: Selector Strings

- Source anchor elements:

```
a:link { color:black }
a:visited { color:yellow }
a:hover { color:green }
a:active { color:red }
```

- Element types that are descendants:

```
ul ol li { letter-spacing:1em }
```

rule applies to `li` element that is part of the content of an `ol` element

CSS Syntax: Selector Strings

- Source anchor elements:

```
a:link { color:black }
a:visited { color:yellow }
a:hover { color:green }
a:active { color:red }
```

- Element types that are descendants:

```
ul ol li { letter-spacing:1em }
```

rule applies to `li` element that is
part of the content of an `ol` element
that is part of the content of a `ul` element

CSS Syntax

- Style rules covered thus far follow ruleset syntax
- At-rule is a second type of rule

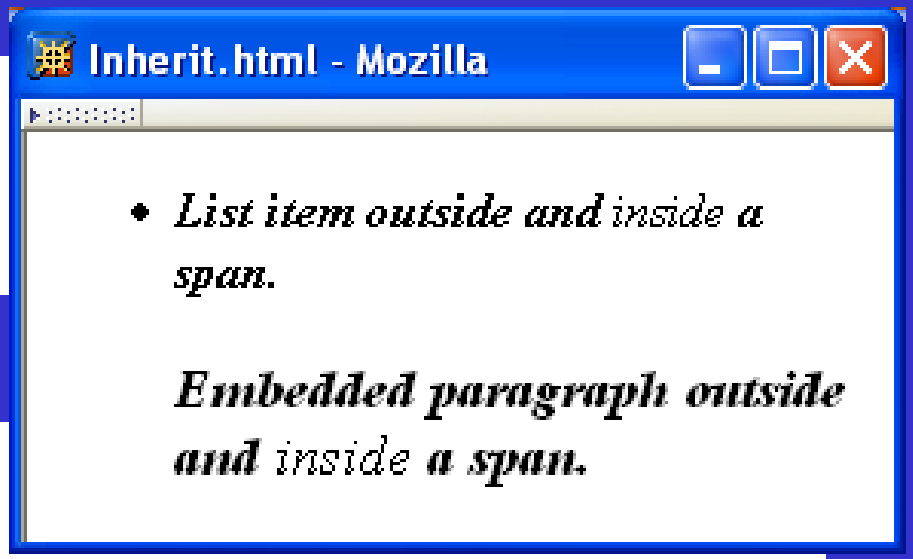
URL relative to style sheet URL

```
@import url("general-rules.css");
```

  - Reads style rules from specified URL
  - Must appear at beginning of style sheet

CSS Inheritance

```
body { font-weight:bold }
li { font-style:italic }
p { font-size:larger }
span { font-weight:normal }
```




```
<body>
  <ul>
    <li>
      List item outside and <span>inside</span> a span.
      <p>
        Embedded paragraph outside and <span>inside</span> a span.
      </p>
    </li>
  </ul>
</body>
```

Selectors

- 3 types of selectors
  - Tag selectors
  - ID selectors
  - Class selectors
- Selectors should never start with a number, nor should they have spaces in them

Tag Selectors

- Can be any type of HTML element
  - body
  - p
  - div

```
p {
background-color: red;
}
```

{ The class Selector

- With the class selector you can define different styles for the same type of HTML element.

```
p.right {text-align: right}
p.center {text-align: center}
```

- Using the class argument in (X)HTML:

```
<p class="right"> This paragraph will be
  right-aligned. </p>
<p class="center"> This paragraph will be
  center-aligned. </p>
```

The class Selector

defines different styles for the same element:

p. center {text-align: center}

in HTML document:

* Paragraph *

The class Selector

.center {text-align: center}

means:

<h1 class="center">Heading is centered</h1>

* Paragraph is centered *

The class Selector

- **Class Selectors**

  <H1 CLASS="warning">Danger!</H1>

  <P CLASS="warning">Be careful…</P>

  …….

  In your HTML code -

  H1.warning {color: red;}

  OR to an entire class…

  .warning {color:red;}

{ Text color

```
<html><head>
<style type="text/css">
   h1 {color: green}
   h2 {color: #dda0dd}
   p {color: rgb(0,0,255)}
</style>
</head>
<body>
   <h1>This is header
   1</h1>
   <h2>This is header
   2</h2>
   <p>This is a
   paragraph</p>
</body>
```

This is header 1

This is header 2

This is a paragraph

ID Attribute

there can be only one element with a given **id** in a document:

- to match all elements with a id:
  - *#intro {font-weight:bold; color: #0000ff}*
- only elements with id="intro":
  - *p#intro {font- weight:bold; color:#0000ff;}*

ID Selectors

- ID's vs. Classes... an epic battle.
- You can not have two elements on one page with the same ID

```
#sidebar {
background-color: blue;
}
```

Grouping Selectors

- You can group selectors together as well with commas

```
p.warning, p.special {
   color: red;
}
```

Now both classes, warning and special, will have red text.

Reset the Styling

```
/* Normalizes margin, padding */
bodv, h1, h2, h3, h4, h5, h6, p, ol, ul, dt, dl, dd, fo,
    blockquote, fieldset, input { padding: 0; margin:
    0; }

/* Normalizes font-size for headers */
      h2, h3, h4, h5, h6 { font-size: 1em; }

/* Removes list-style from lists */
ol, ul { list-style: none; }

/* Removes text-decoration from links */
a { text-decoration: none; }

/* Removes border from img */
a img { border: none; }
```

{ Grouping

```
h1,h2,h3,h4,h5,h6
{
color: green
}
```

- All header elements will be displayed in green text color

This is header h1

This is header h2

This is header h3

This is header h4

Note:

each selector has been separated with a comma.

A Very Brief Overview of Visual Formatting With CSS

- Visual Formatting
  - Fonts
  - Colours
  - Position
  - Box model and Borders

CSS Font Properties

- A font family is a collection of related fonts (typically differ in size, weight, etc.)

```
<p style="font-family:'Jenkins v2.0'">
```

- font-family property can accept a list of families, including generic font families

```
font-family:"Edwardian Script ITC","French Script MT",cursive
```

first choice font

CSS Font Properties

- A font family is a collection of related fonts (typically differ in size, weight, etc.)

```
<p style="font-family:'Jenkins v2.0'">
```

- font-family property can accept a list of families, including generic font families

```
font-family:"Edwardian Script ITC","French Script MT",cursive
```

second choice font

CSS Font Properties

- A font family is a collection of related fonts (typically differ in size, weight, etc.)

```
<p style="font-family:'Jenkins v2.0'">
```

- font-family property can accept a list of families, including generic font families

```
font-family:"Edwardian Script ITC","French Script MT",cursive
```

generic

CSS Font Properties

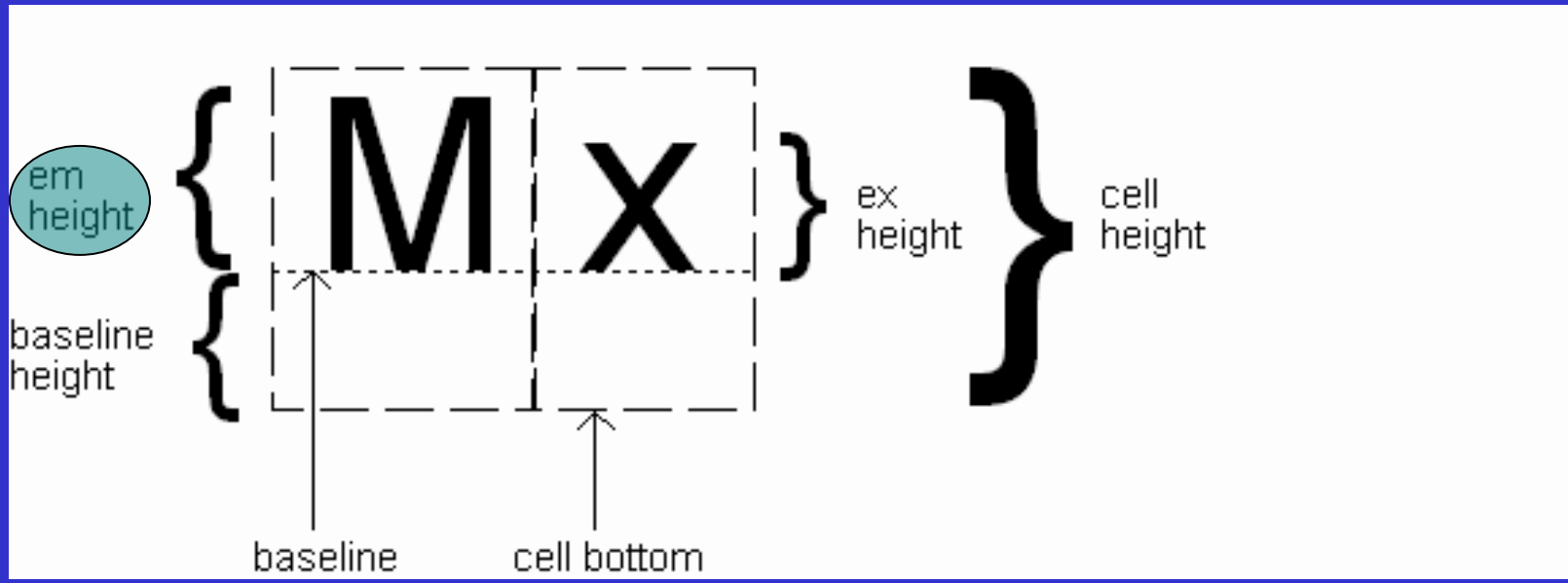
- Many properties, such as `font-size`, have a value that is a CSS length
- All CSS length values except 0 need units

TABLE 3.4: CSS length unit identifiers.

Identifier	Meaning
`in`	inches
`cm`	centimeters
`mm`	millimeters
`pt`	points: 1/72-inch
`pc`	picas: 12 points
`px`	pixel: typically 1/96-inch (see text).
`em`	1em is roughly the height of a capital letter in the reference font (see text).
`ex`	`1ex` is roughly the height of the lowercase 'x' character in the reference font (see text).

CSS Font Properties

Computed value of font-size property



CSS Font Properties

defines em and ex units

- Normally, reference font is the font of the element being styled
- Exception: Using em/ex to specify value for `font-size`

```
<div id="d1" style="font-size:12pt">
  <div id="d2" style="font-size:2em">
```

parent element's font is reference font

CSS Font Properties

- Other ways to specify value for `font-size`:
  - Percentage (of parent `font-size`)

    ```
    font-size:85%
    ```

  - Absolute size keyword: `xx-small`, `x-small`, `small`, `medium` (initial value), `large`, `x-large`, `xx-large`
    - User agent specific; should differ by ~ 20%
  - Relative size keyword: `smaller`, `larger`
    - Relative to parent element's font

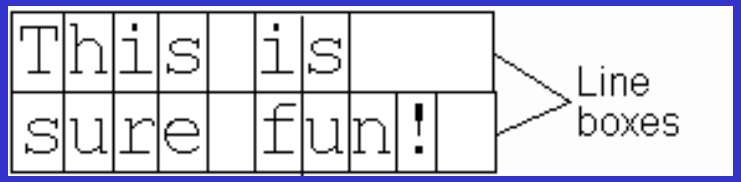
CSS Font Properties

TABLE 3.5: Additional font style properties.

Property	Possible values
`font-style`	`normal` (initial value), `italic` (more cursive than normal), or `oblique` (more slanted than normal).
`font-weight`	`bold` or `normal` (initial value) are standard values, although other values can be used with font families having multiple gradations of boldness (see CSS2 [W3C-CSS-2.0] for details).
`font-variant`	`small-caps`, which displays lowercase characters using uppercase glyphs (small uppercase glyphs if possible), or `normal` (initial value)

CSS Font Properties

- Text is rendered using line boxes

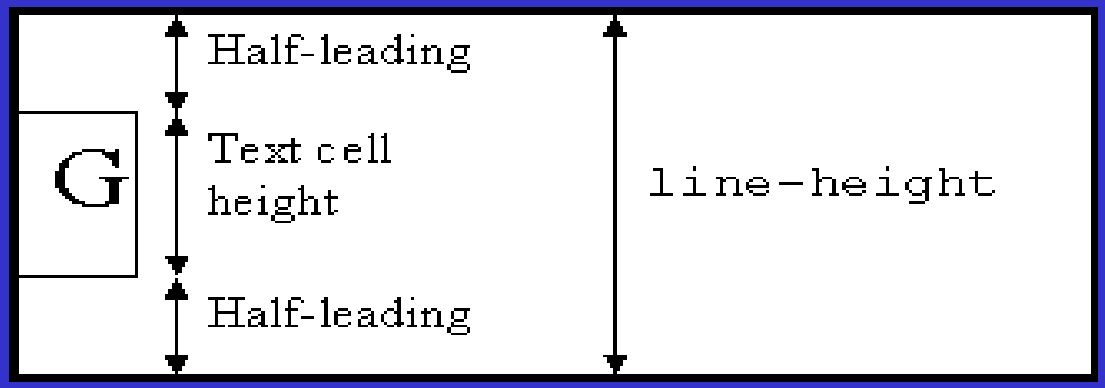


- Height of line box given by
  - Initial value: `normal` (*i.e.*, cell height; relationship with em height is font-specific)
  - Other values (following are equivalent):

```
line-height:1.5em
line-height:150%
line-height:1.5
```

CSS Font Properties

- When `line-height` is greater than cell height:



- Inheritance of `line-height`:
  - Specified value if `normal` or unit-less number
  - Computed value otherwise

CSS Font Properties

- `font` shortcut property:

```
{ font: italic bold 12pt "Helvetica",sans-serif }
```



```
{ font-style: italic;
  font-variant: normal;
  font-weight: bold;
  font-size: 12pt;
  line-height: normal;
  font-family: "Helvetica",sans-serif }
```

CSS Font Properties

- `font` shortcut property:

```
{ font: italic bold 12pt "Helvetica",sans-serif }
```



```
{ font-style: italic;
  font-variant: normal;
  font-weight: bold;
  font-size: 12pt;
  line-height: normal;
  font-family: "Helvetica",sans-serif }
```

Initial values used if no value specified in `font` property list (that is, potentially reset)

CSS Font Properties

- `font` shortcut property:

```
{ font: italic bold 12pt "Helvetica",sans-serif }
```



```
{ font-style: italic;
  font-variant: normal;
  font-weight: bold;
  font-size: 12pt;
  line-height: normal;
  font-family: "Helvetica",sans-serif }
```

specifying line-height (here, twice cell height)

```
{ font: bold oblique small-caps 12pt/2 "Times New Roman",serif }
```

any order

size and family required, order-dependent

Colors

There are sixteen pre-defined colors in HTML:

aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, red, silver, teal, white, and yellow

Which means you can just type the word “blue” into your code, and the browser will give you the default color for blue

Colors

If you want more variety or nuance, there are also 216 “web safe colors” that all browsers should be able to read accurately.

These are the “HEX” color code system, i.e.

#000000 = black (total absence of color)

#FFFFFF = white (presence of all colors)

#FF0000 = red (the red section at full level)

#FF8800 = a shade of orange (red plus green)

Colors

```
Body
    {
    color: black;
    background: white;
    }
```

= text color is black, background color is white

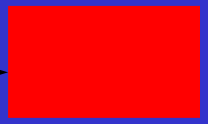
Color Name	RGB Triplet	Hexadecimal	Color Name	RGB Triplet	Hexadecimal
Aqua	(0, 255, 255)	00FFFF	Olive	(128, 128, 0)	808000
Black	(0, 0, 0)	000000	Orange	(255, 165, 0)	FFA500
Blue	(0, 0, 255)	0000FF	Purple	(128, 0, 128)	800080
Fuchsia	(255, 0, 255)	FF00FF	Red	(255, 0, 0)	FF0000
Gray	(128, 128, 128)	808080	Silver	(192, 192, 192)	C0C0C0
Green	(0, 128, 0)	008000	Teal	(0, 128, 128)	008080
Lime	(0, 255, 0)	00FF00	White	(255, 255, 255)	FFFFFF
Maroon	(128, 0, 0)	800000	Yellow	(255, 255, 0)	FFFF00
Navy	(0, 0, 128)	000080			

CSS Color Specifiers

- Predefined names:

```
#ff0000
```

R G B



- 8-bit hexadecimal intensities for red, green, blue:

```
rgb(255,255,0)
```

R G B



- 0-255 decimal intensities:

```
rgb(80%,80%,100%)
```

R G B



Common CSS Properties

Table 8-1 Common CSS properties

CSS Property	Description	Code Example
`background-color`	Specifies a background color for the document	`background-color: blue`
`color`	Specifies the text color of an element	`color: blue`
`font-family`	Specifies the font name	`font-family: arial`
`font-style`	Specifies the font style: normal, italic, or oblique	`font-style: italic`
`margin-left`	Adjusts the left margin	`margin-left: 2in`
`margin-right`	Adjusts the right margin	`margin-right: 2in`
`text-align`	Determines the alignment of text: center, justify, left, or right	`text-align: center`

Common CSS Properties

Table 8-1 Common CSS properties

CSS Property	Description	Code Example
`background-color`	Specifies a background color for the document	`background-color: blue`
`color`	Specifies the text color of an element	`color: blue`
`font-family`	Specifies the font name	`font-family: arial`
`font-style`	Specifies the font style: normal, italic, or oblique	`font-style: italic`
`margin-left`	Adjusts the left margin	`margin-left: 2in`
`margin-right`	Adjusts the right margin	`margin-right: 2in`
`text-align`	Determines the alignment of text: center, justify, left, or right	`text-align: center`

CSS Text Formatting

TABLE 3.6: Primary CSS text properties.

Property	Values
`text-decoration`	`none` (initial value), `underline`, `overline`, `line-through`, or space-separated list of values other than `none`.
`letter-spacing`	`normal` (initial value) or a length representing additional space to be included between adjacent letters in words. Negative value indicates space to be removed.
`word-spacing`	`normal` (initial value) or a length representing additional space to be included between adjacent words. Negative value indicates space to be removed.
`text-transform`	`none` (initial value), `capitalize` (capitalizes first letter of each word), `uppercase` (converts all text to uppercase), `lowercase` (converts all text to lowercase).
`text-indent`	length (initial value 0) or percentage of box width, possibly negative. Specify for block elements and table cells to indent text within first line box.
`text-align`	`left` (initial value for left-to-right contexts), `right`, `center`, or `justified`. Specify for block elements and table cells.
`white-space`	`normal` (initial value), `pre`. Use to indicate whether or not white space should be retained.

CSS Text Color

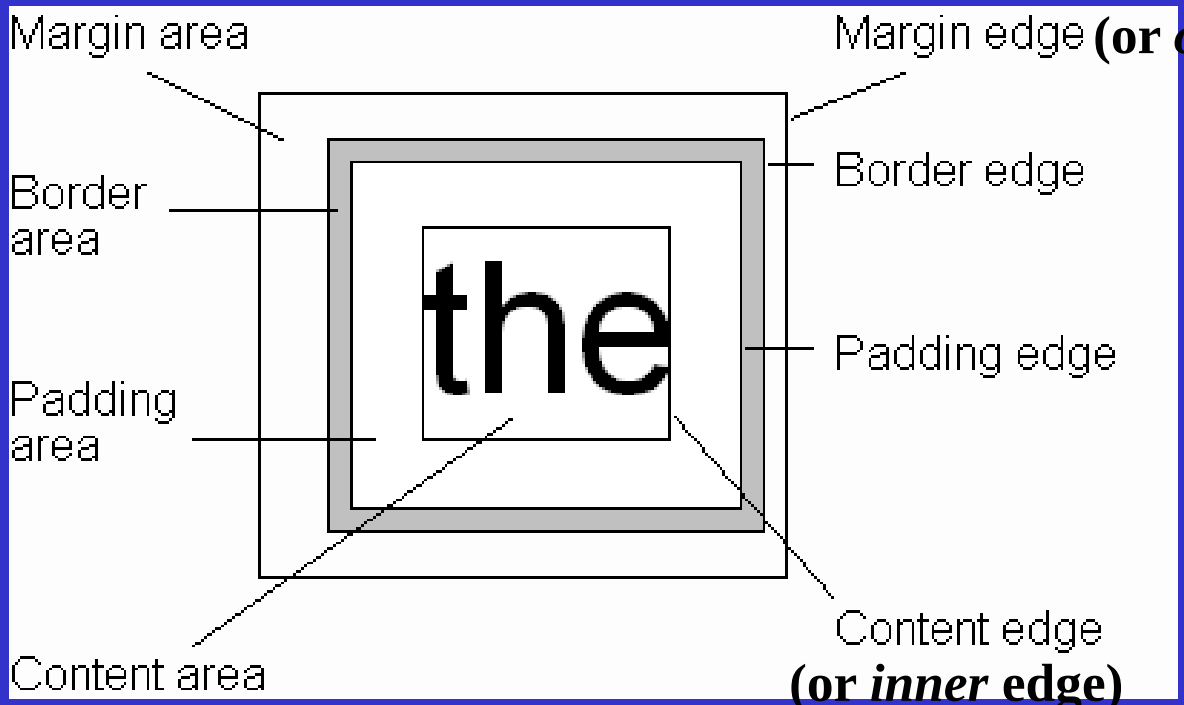
TABLE 3.7: Alternative formats for specifying numeric color values.

Format	Example	Meaning
Functional, integer arguments	`rgb(255,170,0)`	Use arguments as RGB values.
Functional, percentage arguments	`rgb(100%,66.7%,0%)`	Multiply arguments by 255 and round to obtain RGB values (at most one decimal place allowed in arguments).
Hexadecimal	`#ffaa00`	The first pair of hexadecimal digits represents the red intensity, second and third represent green and blue, respectively.
Abbreviated hexadecimal	`#fa0`	Duplicate the first hexadecimal digit to obtain red intensity, duplicate second and third to obtain green and blue, respectively.

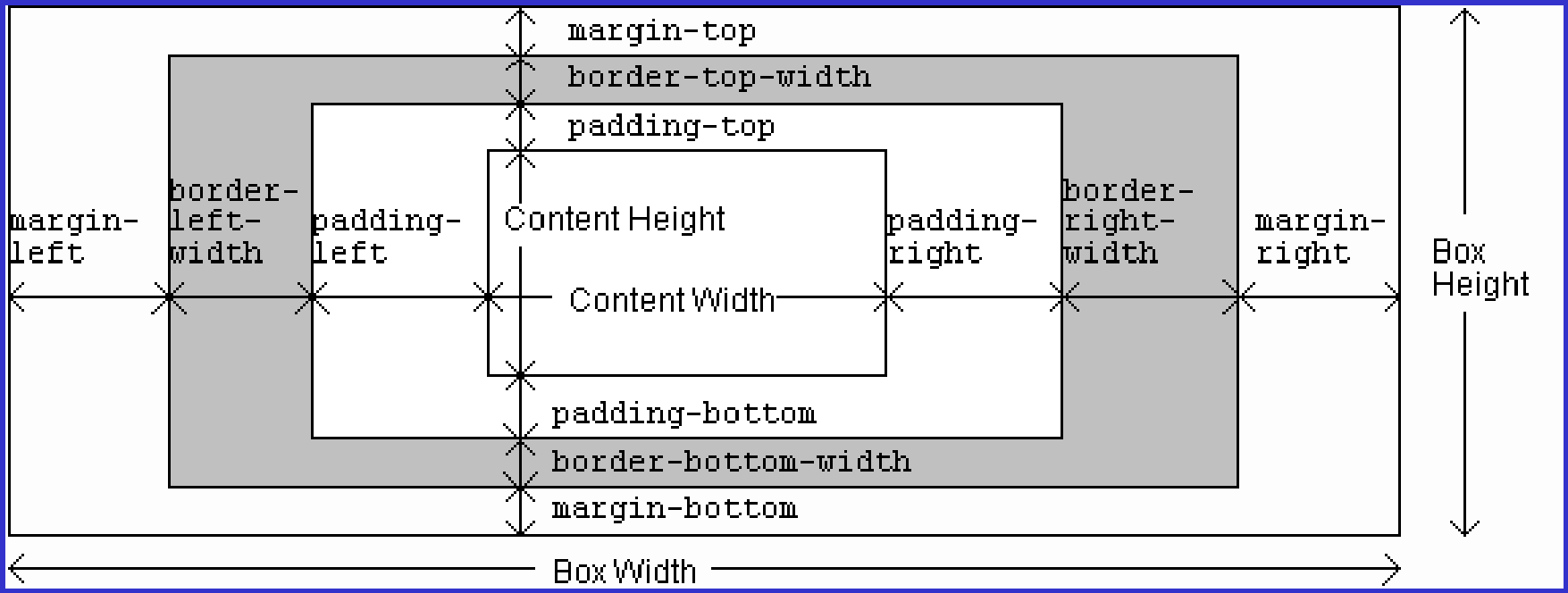
ased on Jeffrey C. Jackson's slides

CSS Box Model

- Every rendered element occupies a box:



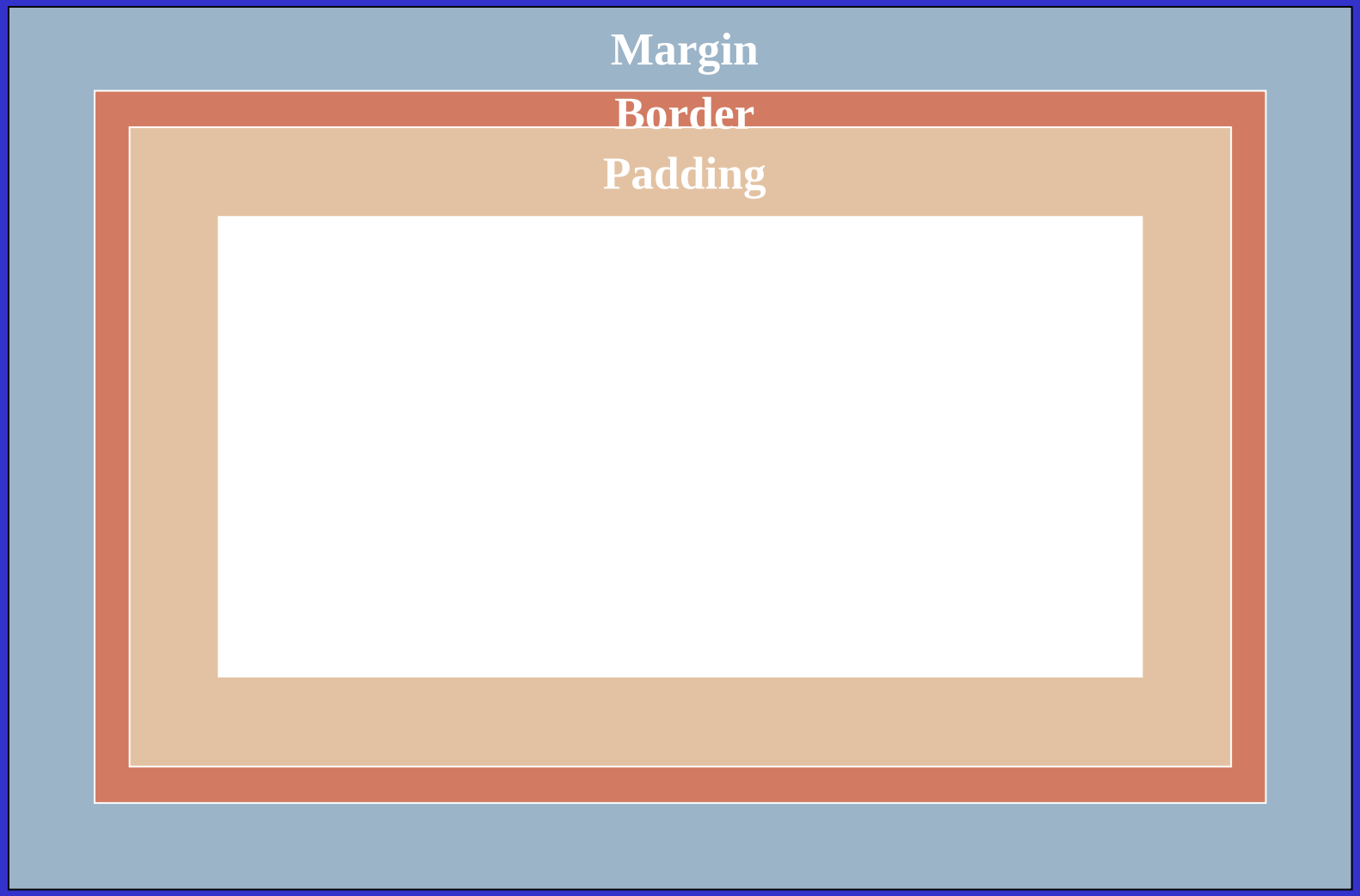
CSS Box Model



The CSS Box Model

- Every CSS box is divided into regions, consisting of:
    1. Content
    2. Padding
    3. Border
    4. Margins

The CSS Box Model



Margins & Padding

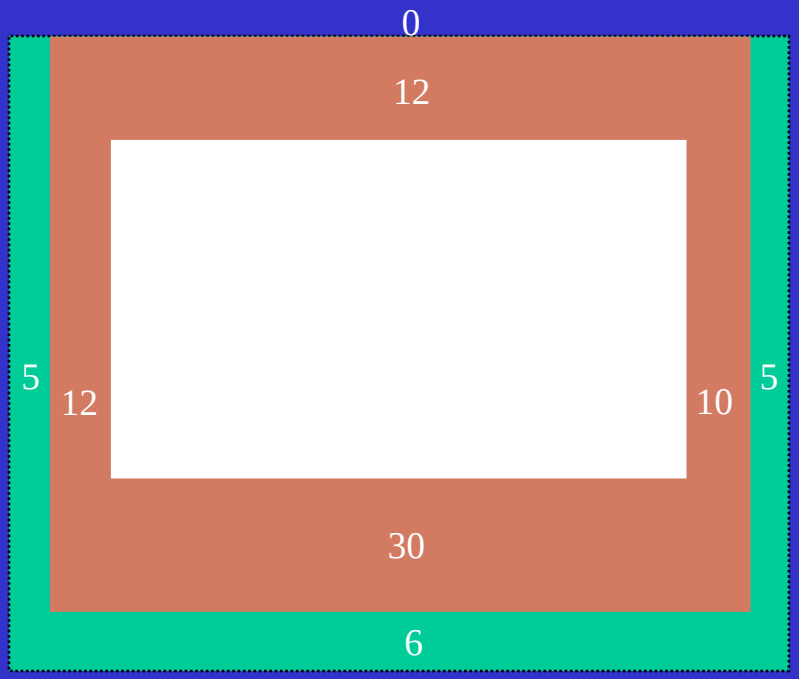
- **Margins**
- Margins define the space around elements outside the border
- Margin properties can have negative values in order to deliberately overlap content
- Margin properties will affect the position of background elements (graphics and/or colours) in relation to the edges of the containing block element
- Margin properties can be defined independently on top, right, bottom and left, or all-at-once using CSS shorthand

Margins & Padding

- **Padding**
- Padding defines the space around elements inside the border; i.e between the border and the content itself
- Padding properties cannot have negative values
- Padding properties do not affect the position of background elements (graphics and/or colours) in the containing block element; only the position of content.
- Padding properties can be defined independently on top, right, bottom and left, or all-at-once using CSS shorthand

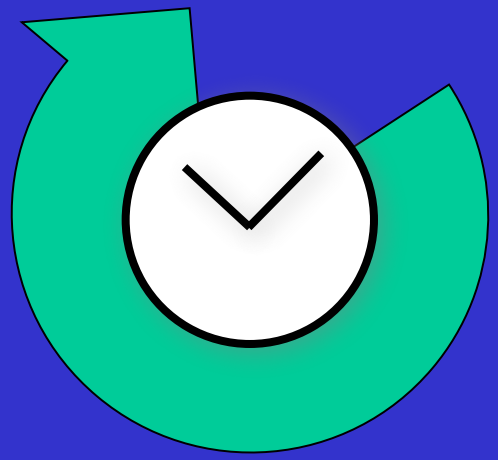
CSS Shorthand: Margin & Padding

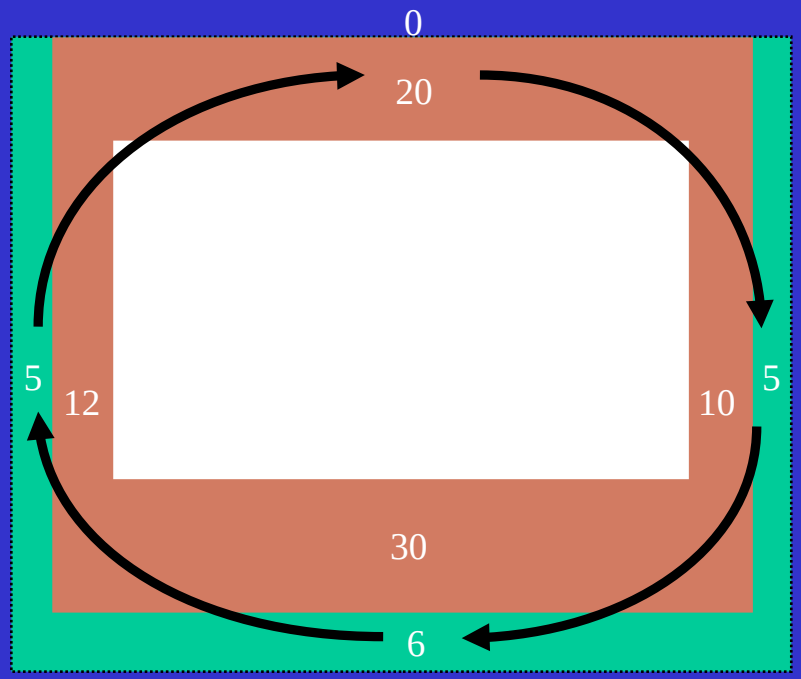
- For margin and padding (and others), CSS provides a number of shorthand properties that can save on writing lines and lines of code. Instead of writing this:
- #container {
  margin-top: 0;
  margin-right: 5px;
  margin-bottom: 6px;
  margin-left: 5px;
  padding-top: 20px;
  padding-right: 10px;
  padding-bottom: 30px;
  padding-left: 12px;
  }



CSS Shorthand: Margin & Padding

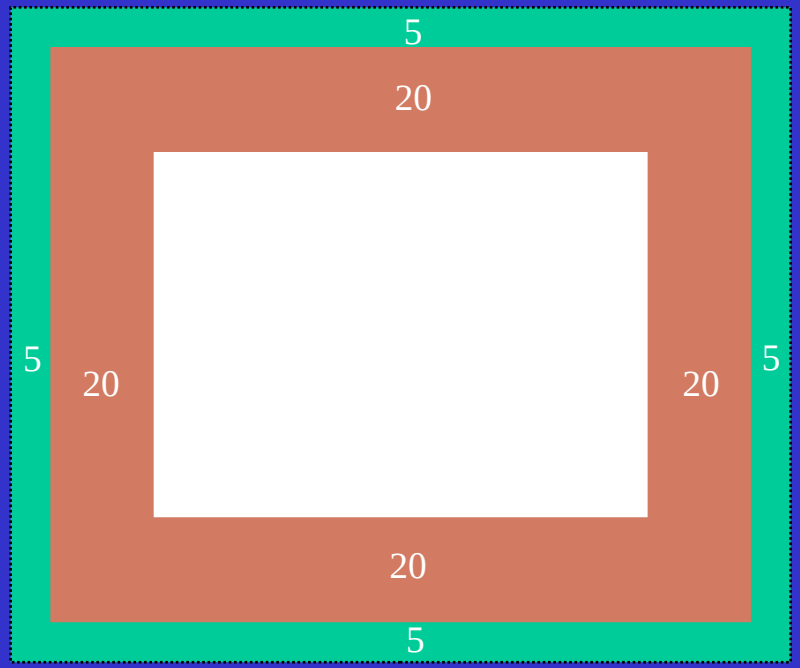
- …Its much easier to write this:
- #container {
  padding: 20px 10px 30px 12px;
  margin: 0px 5px 6px 5px;
  }
- The sequence order is always clockwise, starting from the top





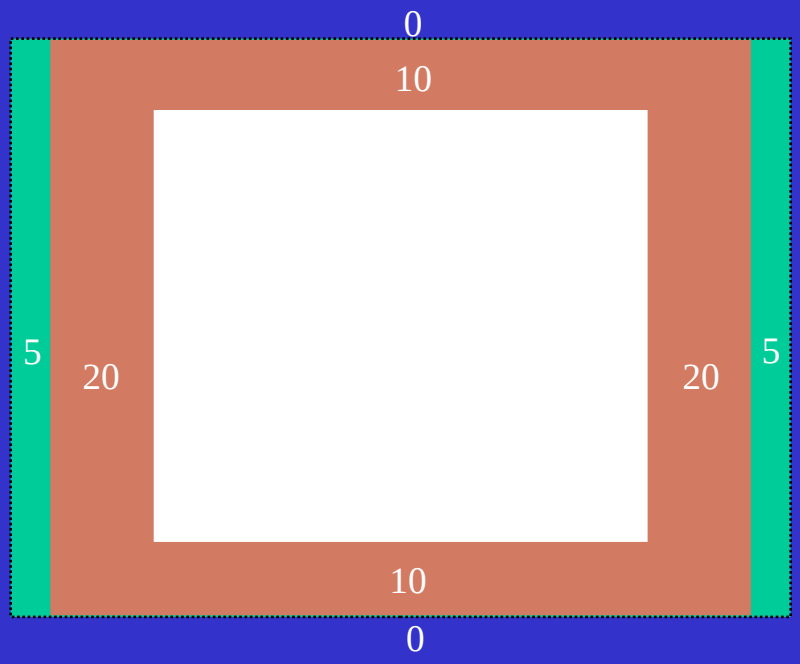
CSS Shorthand: Margin and Padding

- You can also apply just <u>one</u> value, example:
- #container {
  padding: 20px;
  margin: 5px;
  }
- Which will apply the value specified equally on all 4 sides



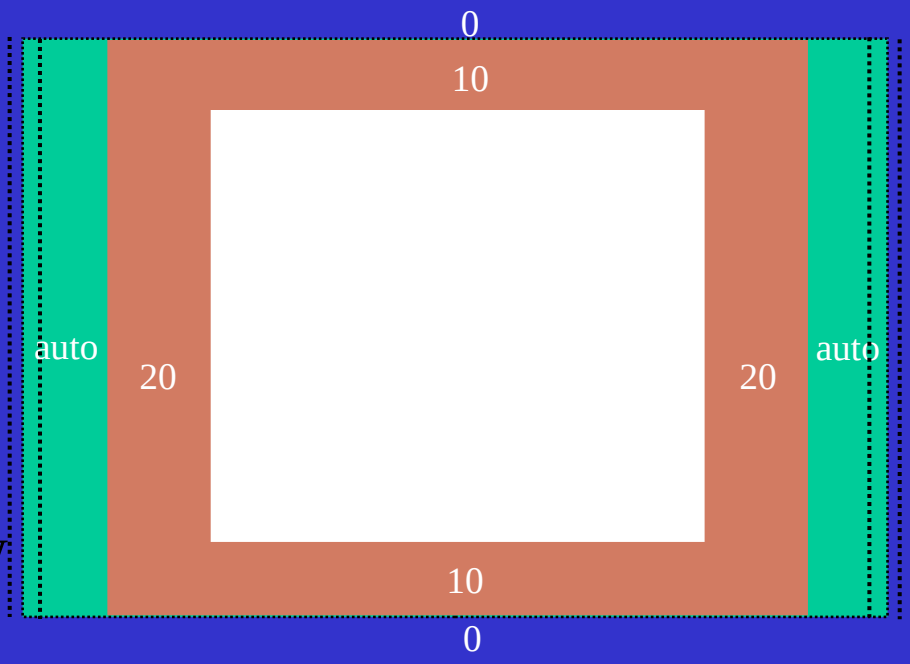
CSS Shorthand: Margin and Padding

- And you can apply <u>two</u> values, example:
- #container {
  padding: 10px 20px;
  margin: 0px 5px;
  }
- The first value is applied to the top and bottom
- The second value is applied to the left and right



CSS Shorthand: Margin and Padding: auto

- A useful value to remember is ‘auto’:
- #container {
  padding: 10px 20px;
  margin: 0px auto;
  }
- Usually applied to the left & right areas of the margin property, auto is useful for centering a block container element in the browser window



{ Background:

- Control over the background color of an element
- set an image as the background,
- repeat a background image

- `background-color`
  - *color-rgb*
    color-hex
    color-name
- `background-image`
  - `url(URL)`
    `none`
- `background-repeat`
  - `repeat`
    `repeat-x`
    `repeat-y`
    `no-repeat`

```
<html>
<head>

<style type="text/css">
body
{
background-image:
url('bgdesert.jpg');
background-repeat: repeat-x
}
</style>

</head>

<body>
</body>
</html>
```



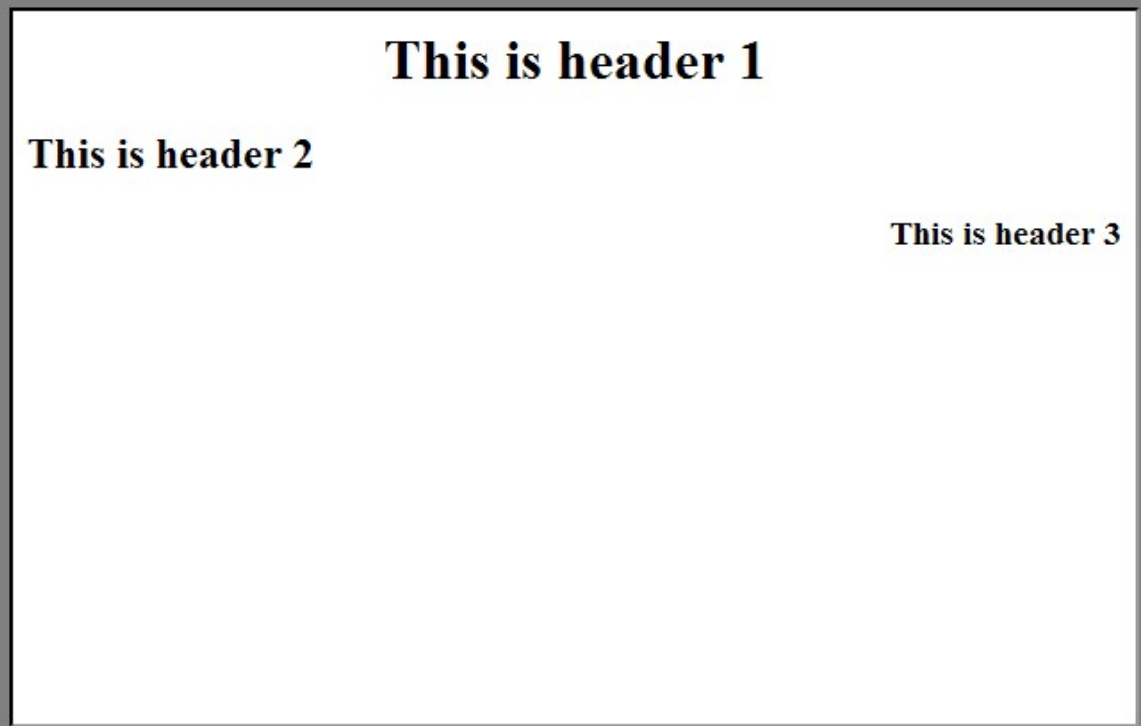
{Text:

- color
- direction
  - ltr
    rtl
- word spacing
  - normal
    length
- text-decoration
  - none
    underline
    overline
    line-through
    blink
- text-align
  - left
    right
    center
    justify

```
<html>
<head>
<style type="text/css">
h1 {text-align: center}
h2 {text-align: left}
h3 {text-align: right}
</style>
</head>

<body>
<h1>This is header 1</h1>
<h2>This is header 2</h2>
<h3>This is header 3</h3>
</body>

</html>
```



{Font:

- font-family
  - *family-name*
    generic-family
- font-size
  - xx-small
    x-small
    small
  - /etc./
- font-weigh
  - normal
    bold
    bolder
    lighter
    100

{Dimension

```
img.normal
{
height: auto
}

img.big
{
height: 160px
}

img.small
{
height: 30px
}
</style>
</head>
<body>

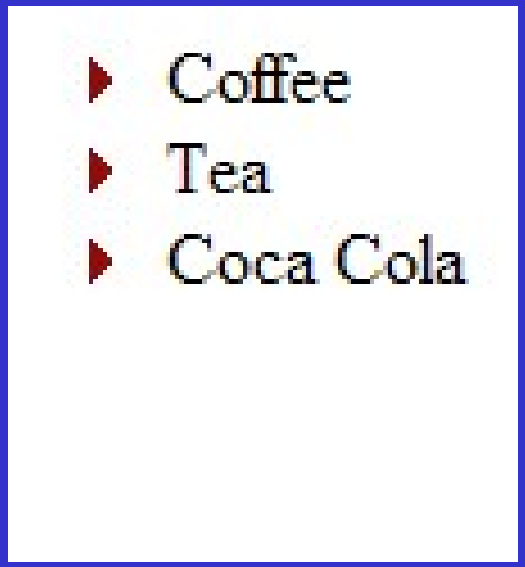
<img class="normal" src="logocss.gif" width="95" height="84" />
<br />
<img class="big" src="logocss.gif" width="95" height="84" />
<br />
<img class="small" src="logocss.gif" width="95" height="84" />
```



{List:

```
<head>
<style type="text/css">
  ul
  {
  list-style-image:
    url('arrow.gif')
  }
</style>
</head>

<body>
  <ul>
   <li>Coffee</li>
   <li>Tea</li>
   <li>Coca Cola</li>
  </ul>
</body>
```



Visual Formatting: Images

- `position`:
  `static`, `relative`, `absolute`, `fixed`
- Static — normal elements
- Relative — translate from usual position
- Absolute — scroll with the page
- Fixed — like absolute, but don't scroll away
- Example: Jon Gunderson

Visual Formatting: Images (cont.)

- `z-index`: depth

- `float` and `clear`
  - `float: left` or `float: right` or `float: none`
    Position relative to parent element
  - Reset with clear

```
<br style="clear:both" />
```

Basic Formatting of the ‘Jump Box’

HTML Outline

```
<body>
   <!-- … -->
   <div class="up">
      <dl>
         <dt>Jump to
           top</dt>
      <!-- … -->
   </div>
</body>
```

Extract of CSS Rules

```
body
   {margin-left: 6.3em}
div.up
   {position: fixed;
   left: 1em;
   top: 40%;
   padding: .2ex;
   min-width: 5.5ex }
```

Relative and Absolute Positioning

- **position** property:
  - static – box is positioned according to normal flow
  - relative – box is positioned relative to where it would have been placed in the normal flow
  - absolute – box is positioned relative to its containing block; these boxes are taken out of flow
  - fixed – box is fixed in relation to the viewport (a browser window or a printed page)

Margin Styles

- Control your margins with the following four styles:
  - **margin-top:** ***length***
  - **margin-right:** ***length***
  - **margin-bottom:** ***length***
  - **margin-left:** ***length***
- Margin values can also be negative. This creates an overlay effect by forcing the browser to render one element on top of another
- You can also combine the four margin styles into a single style:
  - ***margin: top right bottom left***

Padding Styles

- Styles to set padding are similar to styles to set margins:
  - `padding-top: value`
  - `padding-right: value`
  - `padding-bottom: value`
  - `padding-left: value`

Border Styles

- border-top-width: *length*
- border-right-width: *length*
- border-bottom-width: *length*
- border-left-width: *length*
- border-width: *top right bottom left*
- border-top-color: *color*
- border-right-color: *color*
- border-bottom-color: *color*
- border-left-color: *color*
- border-color: *top right bottom left*

Border Styles

- border-top-style: *type*
- border-right-style: *type*
- border-bottom-style: *type*
- border-left-style: *type*
- border-style: *top right bottom left*

Table Design

- Tables can build a nice layout
- Tables are predictable

Alternative Design Method

- DIVs can be an alternate to `<table>`
- DIVs are a container like a table cell
- CSS can position the DIV

```
<div id="article">xxx</div>
#article{
  width:250px;
  padding:5px;
  float:right;}
```

DIV Design

Use DIVs to create the skeleton of the page.

- There should be no display-specific information in the XHTML
- ***The Goal***: separate the information from the layout / presentation of the page
- Layout is entirely controlled by CSS

File Edit Search Project View Format Column Macro Advanced Window Help

index.html Edit1*

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
<meta http-equiv="content-type" content="text/html; charset=iso-8859-1" />
<title>Style3</title>
<link rel="stylesheet" href="format.css" type="text/css" media="screen" />
</head>

<body>

   <div id="oldbrowser">
      <p>This site will look much better in a browser that supports <a href="http://www.webstandards.org/upgrade/" title="Download a browser that complies
   </div>

   <div id="overall">
      <a href='#content' id='skipLink'>Skip to content</a>
      <div id="masthead"><span>This is a title for the page</span></div>
      <div id="divider"><hr /></div>

      <div id="searchbar">
         <div id="search">Search</div>
         <div id="quicklinks">List of quick links</div>
      </div>

      <div id="menu">
         <ul>
            <li><a href='index.html'>Menu</a></li>
            <li><a href='index.html'>Menu</a></li>
            <li><a href='index.html'>Menu</a></li>
            <li><a href='index.html'>Menu</a></li>
            <li><a href='index.html'>Menu</a></li>
            <li><a href='index.html'>Menu</a></li>
         </ul>
      </div>

      <div id="content">
         <div id='contentBuffer' style='width: 99%;'>
         Test
         </div>
      </div>

      <div id="bottom"><span>UWP...What College Should Be</span></div>
      <div id="divider2"><hr /></div>
      <div id="footer">
         Copyright Information
      </div>
   </div>

</body>
</html>
```

For Help, press F1 | Ln 20, Col. 27, C0 | DOS | Mod: 9/24/2003 1:27:31PM | File Size: 1580 | INS

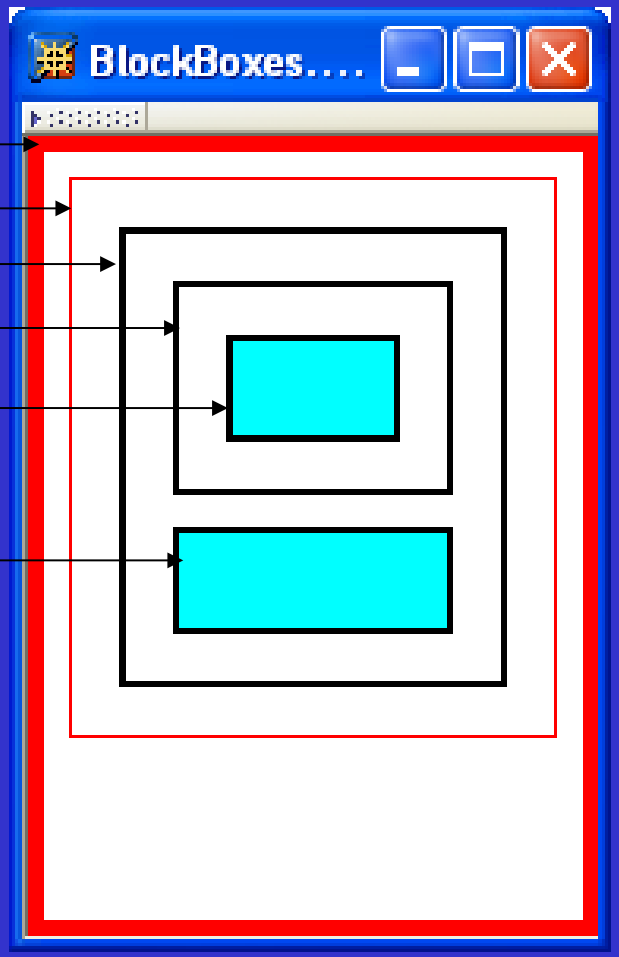
Normal Flow Layout

```
html, body { border:solid red thin }
html { border-width:thick }
body { padding:15px }
div { margin:0px; padding:15px; border:solid black 2px }
.shade { background-color:aqua }
.topMargin { margin-top:10px }
```

```
<body>
  <div id="d1">
    <div id="d2">
      <div id="d3" class="shade"></div>
    </div>
    <div id="d4" class="shade topMargin"></div>
  </div>
</body>
```

Blo
ck
ele
me

Normal Flow Layout



Top edges of

Building Navigation Bars

- At heart, a navigation bar is nothing more than a bunch of links. More specifically, it's actually a of the different sections of a site. Lists provide us with a way of grouping related elements and, by doing so, we give them meaning and structure.

Building Navigation Bars

- A navigation menu is based on a simple *list* inside a div, like this:

```
div id="NavBar">
<ul>
<li><a href="#">Home</a></li>
<li><a href="#">About</a></li>
<li><a href="#">Events<a/></li>
<li><a href="#">Forum</a></li>
</ul>
</div>
```

Title

- Divs help add structure to a page. Div stands for *division* and marks a logical group of elements on a page.
  - Divs divide the page into rectangular, box-like areas. Invisible unless you turn on borders or color background.
  - Include tags for all major regions of your page, such as header, main content, sidebar, etc.

Divs

- Once your tags are in place, add either a class or ID for styling each separately.
- For parts of the page that appear only once and form the basic building blocks of the page, web designers usually use an ID.
- ID selectors are identified using a hash character (#); class selectors are identified with a period(.).

```
<link rel=File-List href="./Hi%20Mom_files/filelist.xml">
<title>Hi Mom</title>
<!--[if gte mso 9]><xml>
 <o:DocumentProperties>
  <o:Author>siegs</o:Author>
  <o:LastAuthor>siegs</o:LastAuthor>
  <o:Revision>1</o:Revision>
  <o:TotalTime>0</o:TotalTime>
  <o:Created>2003-09-15T17:42:00Z</o:Created>
  <o:LastSaved>2003-09-15T17:42:00Z</o:LastSaved>
  <o:Pages>1</o:Pages>
  <o:Company>UW-Platteville</o:Company>
  <o:Lines>1</o:Lines>
  <o:Paragraphs>1</o:Paragraphs>
  <o:Version>9.3821</o:Version>
 </o:DocumentProperties>
</xml><![endif]-->
<style>
<!--
 /* Style Definitions */
p.MsoNormal, li.MsoNormal, div.MsoNormal
        {mso-style-parent:"";
        margin:0in;
        margin-bottom:.0001pt;
        mso-pagination:widow-orphan;
        font-size:12.0pt;
        font-family:"Times New Roman";
        mso-fareast-font-family:"Times New Roman";}
@page Section1
        {size:8.5in 11.0in;
        margin:1.0in 1.25in 1.0in 1.25in;
        mso-header-margin:.5in;
        mso-footer-margin:.5in;
        mso-paper-source:0;}
div.Section1
        {page:Section1;}
-->
</style>
</head>

<body lang=EN-US style='tab-interval:.5in'>

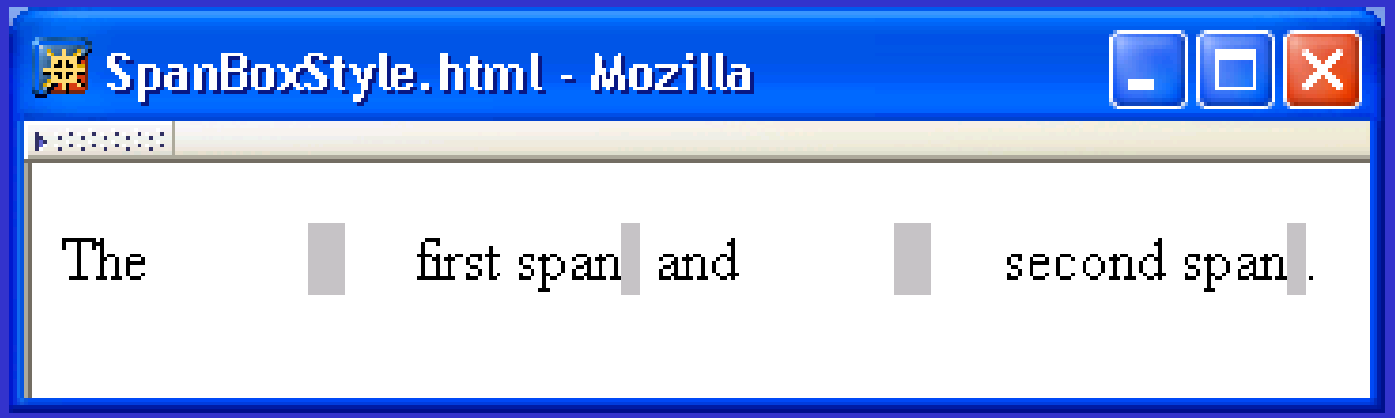
<div class=Section1>

<p class=MsoNormal>Hi Mom</p>

</div>
```

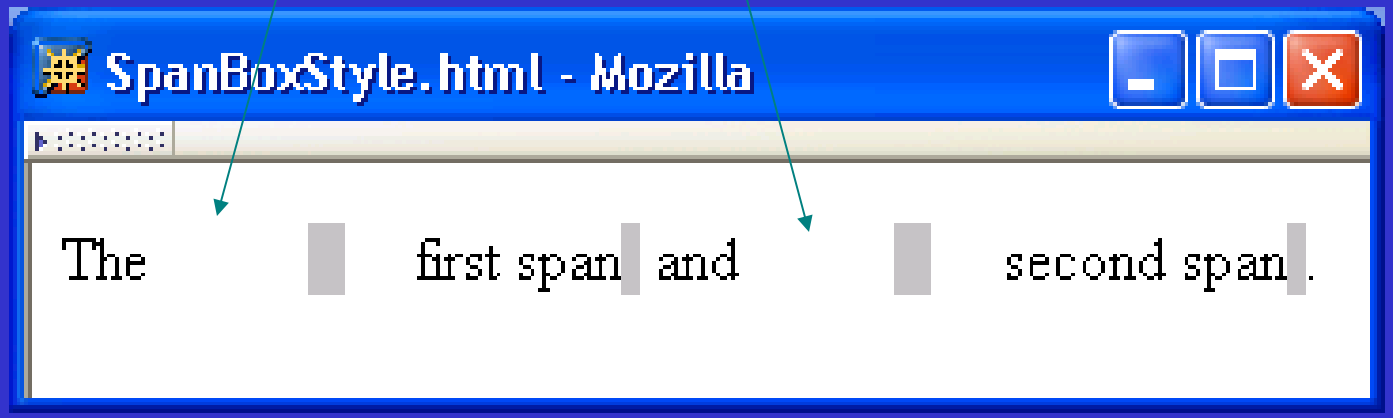
CSS Box Model

```
span { margin-left: 1cm;
       border-left-width: 10px;
       border-left-color: silver;
       border-left-style: solid;
       padding-left: 0.5cm;
       border-right-width: 5px;
       border-right-color: silver;
       border-right-style: solid }
```



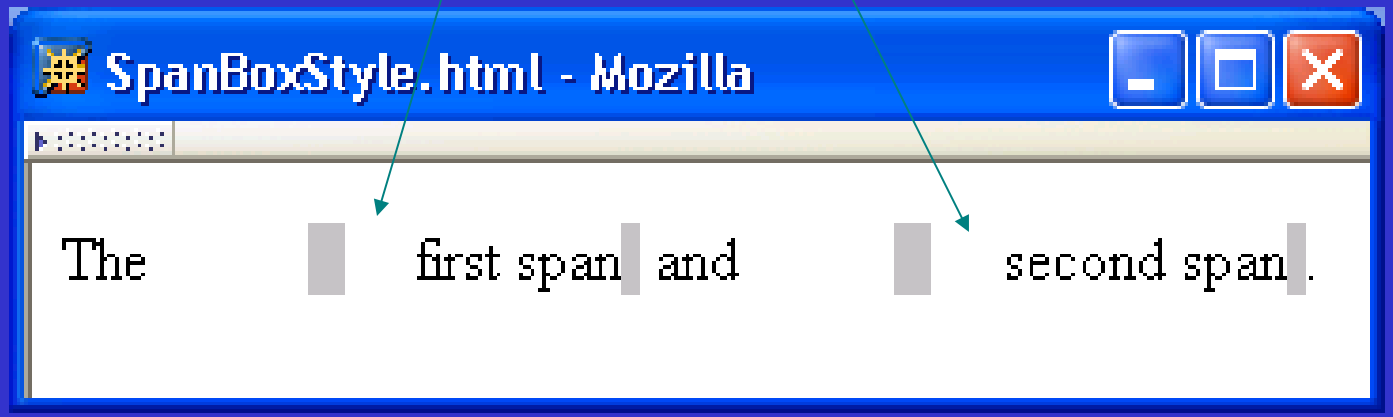
CSS Box Model

```
span { margin-left: 1cm;
       border-left-width: 10px;
       border-left-color: silver;
       border-left-style: solid;
       padding-left: 0.5cm;
       border-right-width: 5px;
       border-right-color: silver;
       border-right-style: solid }
```



CSS Box Model

```
span { margin-left: 1cm;
       border-left-width: 10px;
       border-left-color: silver;
       border-left-style: solid;
       padding-left: 0.5cm;
       border-right-width: 5px;
       border-right-color: silver;
       border-right-style: solid }
```



CSS Box Model

TABLE 3.9: Basic CSS style properties associated with the box model.

Property	Values
`padding-{top,right,bottom,left}`	CSS length (Sec. 3.6.2).
`padding`	One to four length values (see text).

TABLE 3.10: Meaning of values for certain shorthand properties that take one to four values.

Number of values	Meaning
One	Assign this value to all four associated properties (`top`, `right`, `bottom`, and `left`).
Two	Assign first value to associated `top` and `bottom` properties, second value to associated `right` and `left` properties.
Three	Assign first value to associated `top` property, second value to `right` and `left`, and third value to `bottom`.
Four	Assign first value to associated `top` property, second to `right`, third to `bottom`, and fourth to `left`.

CSS Box Model

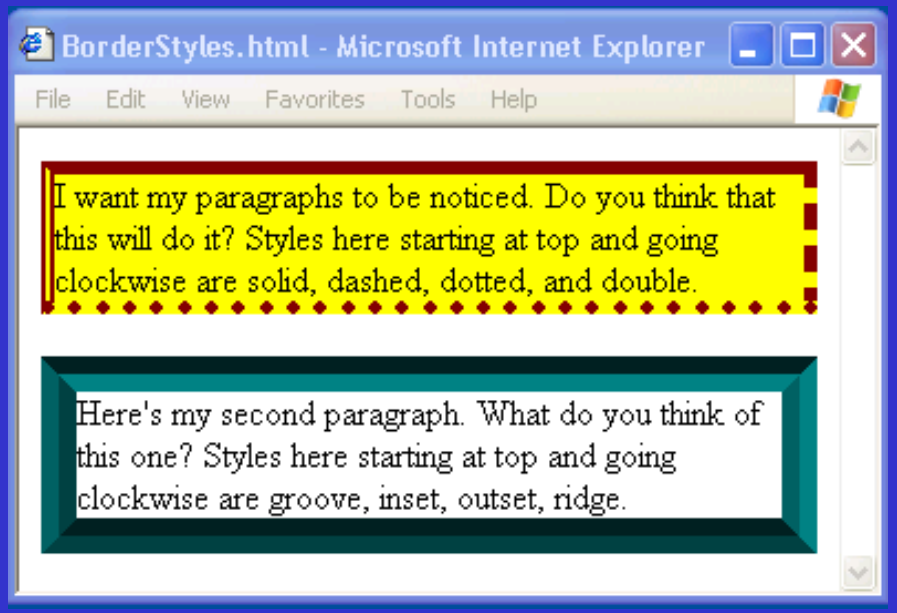
TABLE 3.9: Basic CSS style properties associated with the box model.

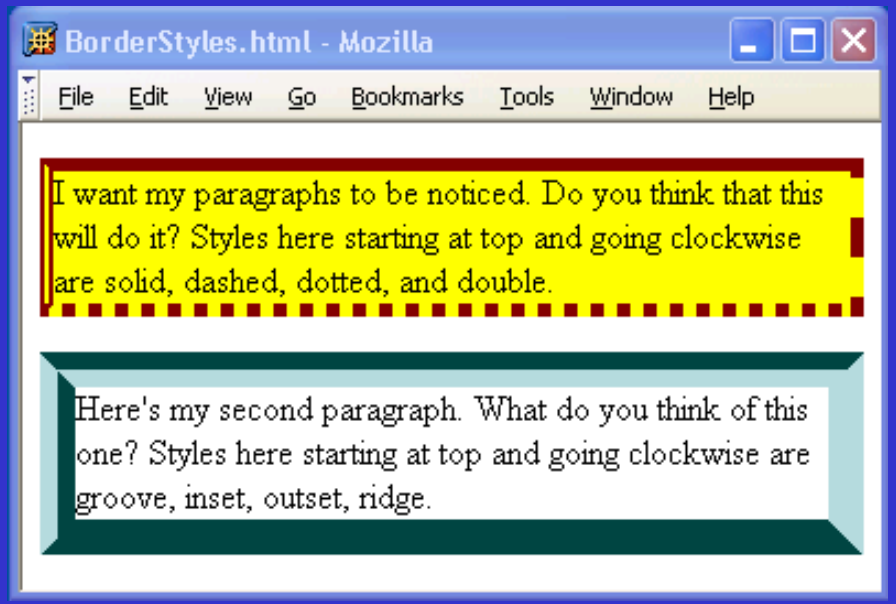
`border-{top,right,bottom,left}-width`	`thin`, `medium` (initial value), `thick`, or a length.
`border-width`	One to four `border-*-width` values.
`border-{top,right,bottom,left}-color`	Color value. Initial value is value of element's `color` property.
`border-color`	`transparent` or one to four `border-*-color` values.

CSS Box Model

TABLE 3.9: Basic CSS style properties associated with the box model.

`border-{top,right,bottom,left}-style`	`none` (initial value), `hidden`, `dotted`, `dashed`, `solid`, `double`, `groove`, `ridge`, `inset`, `outset`.
`border-style`	One to four `border-*-style` values.





CSS Box Model

TABLE 3.9: Basic CSS style properties associated with the box model.

`border-{top,right,bottom,left}`	One to three values (in any order) for `border-*-width`, `border-*-color`, and `border-*-style`. Initial values are used for any unspecified values.
`border`	One to three values; equivalent to specifying given values for each of `border-top`, `border-right`, `border-bottom`, and `border-left`.
`margin-{top,right,bottom,left}`	`auto` (see text) or length.
`margin`	One to four `margin-*` values.

CSS Box Model

- If multiple declarations apply to a property, the last declaration overrides earlier specifications

```
{ border: 15px solid;
  border-left: 30px inset red;
  color: blue }
```

Left border is 30px wide, inset style,

Backgrounds

- `background-color`
  - Specifies background color for content, padding, and border areas
  - Margin area is always transparent
  - Not inherited; initial value `transparent`
- `background-image`
  - Specifies (using `url()` function) image that will be tiled over an element